LAPORAN WORKSHOP DEVELOPER OPERATIONAL KUBERNETES

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PROGRAM STUDI TEKNIK INFORMATIKA

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI JEMBER

2026

1. Install Kubernetes

```
PS C:\> winget install Kubernetes.minikube
The `msstore` source requires that you view the following agreements before using.
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
The source requires the current machine's 2-letter geographic region to be sent to the backend service to function properly (ex. "US").

Do you agree to all the source agreements terms?
[Y] Yes  [N] No: Y
Found Kubernetes - Minikube - A Local Kubernetes Development Environment [Kubernetes.minikube] Version 1.38.1
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://github.com/kubernetes/minikube/releases/download/v1.38.1/minikube-installer.exe
  ██████████████████████████  48.5 MB / 48.5 MB
Successfully verified installer hash
Starting package install...
Successfully installed
```

2. Setup Kubernetes

```
PS C:\> minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 11 Home Single Language 25H2
* Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 2.97 Mi
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 5.02 Mi
* Creating docker container (CPUs=2, Memory=3072MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS C:\> kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   31s   v1.35.1
```

3. Buat folder Project

```
PS C:\> mkdir kubernetes


    Directory: C:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         4/29/2026   8:55 AM                kubernetes


PS C:\> cd kubernetes
PS C:\kubernetes> code .
```

Buat File YAML

a. Deployment.yaml

deployment.yaml ✕ service.yaml

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web-nginx
  template:
    metadata:
      labels:
        app: web-nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

b. Service.yaml

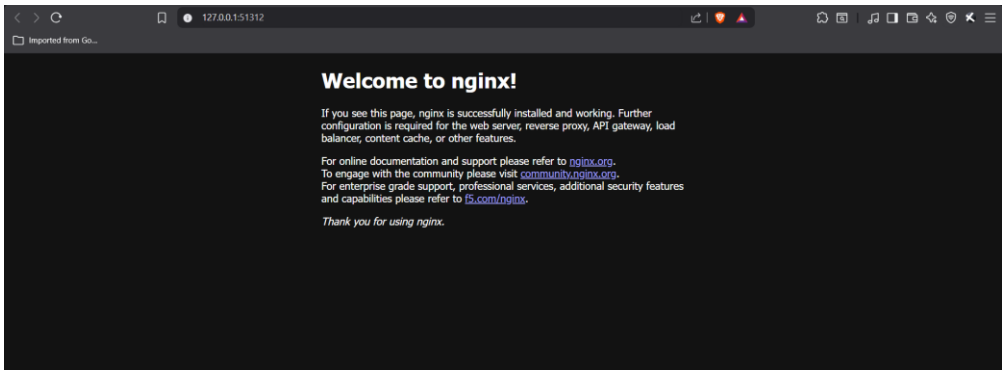
```yaml
apiVersion: v1
kind: Service
metadata:
  name: web-nginx-service
spec:
  selector:
    app: web-nginx
  ports:
    - port: 80
      targetPort: 80
  type: NodePort
```

4. Menjalankan YAML

```
PS C:\kubernetes> kubectl apply -f deployment.yaml
deployment.apps/web-nginx created
PS C:\kubernetes> kubectl apply -f service.yaml
service/web-nginx-service created
PS C:\kubernetes> kubectl get pods
NAME                          READY   STATUS              RESTARTS   AGE
web-nginx-748d764f7-4zvm6     0/1     ContainerCreating   0          20s
web-nginx-748d764f7-fxb8r     0/1     ContainerCreating   0          20s
web-nginx-748d764f7-j2hb9     0/1     ContainerCreating   0          20s
PS C:\kubernetes> kubectl get pods -w
NAME                          READY   STATUS              RESTARTS   AGE
web-nginx-748d764f7-4zvm6     0/1     ContainerCreating   0          35s
web-nginx-748d764f7-fxb8r     0/1     ContainerCreating   0          35s
web-nginx-748d764f7-j2hb9     0/1     ContainerCreating   0          35s
web-nginx-748d764f7-fxb8r     1/1     Running             0          37s
web-nginx-748d764f7-j2hb9     1/1     Running             0          39s
web-nginx-748d764f7-4zvm6     1/1     Running             0          42s
```

5. Membuka Aplikasi

```
PS C:\kubernetes> minikube service web-nginx-service
|-----------|-----------------|-------------|---------------------------|
| NAMESPACE |      NAME       | TARGET PORT |           URL             |
|-----------|-----------------|-------------|---------------------------|
| default   | web-nginx-service | 80          | http://192.168.49.2:31276 |
|-----------|-----------------|-------------|---------------------------|
* Starting tunnel for service web-nginx-service.
|-----------|-----------------|-------------|---------------------|
| NAMESPACE |      NAME       | TARGET PORT |         URL         |
|-----------|-----------------|-------------|---------------------|
| default   | web-nginx-service |             | http://127.0.0.1:51312 |
|-----------|-----------------|-------------|---------------------|
* Opening service default/web-nginx-service in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
* Stopping tunnel for service web-nginx-service.
```



6. Scaling Pod

   a. Terminal 2

```
C:\>kubectl scale deployment web-nginx --replicas=5
deployment.apps/web-nginx scaled

C:\>minikube node add
* Adding node m02 to cluster minikube as [worker]
! Cluster was created without any CNI, adding a node to it might cause broken networking.
* Starting "minikube-m02" worker node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Creating docker container (CPUs=2, Memory=2200MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Verifying Kubernetes components...
* Successfully added m02 to minikube!
```

   b. Terminal 1

```
PS C:\kubernetes> kubectl get pods -w
NAME                        READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-4zvm6   1/1     Running   0          4m58s
web-nginx-748d764f7-fxb8r   1/1     Running   0          4m58s
web-nginx-748d764f7-j2hb9   1/1     Running   0          4m58s
web-nginx-748d764f7-prfdl   0/1     Pending   0          0s
web-nginx-748d764f7-42xqj   0/1     Pending   0          0s
web-nginx-748d764f7-42xqj   0/1     Pending   0          0s
web-nginx-748d764f7-prfdl   0/1     Pending   0          0s
web-nginx-748d764f7-42xqj   0/1     ContainerCreating   0          0s
web-nginx-748d764f7-prfdl   0/1     ContainerCreating   0          0s
web-nginx-748d764f7-prfdl   1/1     Running             0          12s
web-nginx-748d764f7-42xqj   1/1     Running             0          20s
```

7. Menambahkan Worker Node

```
C:\>minikube node add
* Adding node m02 to cluster minikube as [worker]
! Cluster was created without any CNI, adding a node to it might cause broken networking.
* Starting "minikube-m02" worker node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Creating docker container (CPUs=2, Memory=2200MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Verifying Kubernetes components...
* Successfully added m02 to minikube!

C:\>kubectl get nodes
NAME           STATUS     ROLES           AGE   VERSION
minikube       Ready      control-plane   10m   v1.35.1
minikube-m02   NotReady   <none>          19s   v1.35.1

C:\>minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

minikube-m02
type: Worker
host: Running
kubelet: Running


C:\>kubectl get pods -o wide
NAME                        READY   STATUS    RESTARTS   AGE    IP           NODE       NOMINATED NODE   READINESS GATES
web-nginx-748d764f7-42xqj   1/1     Running   0          3m     10.244.0.6   minikube   <none>           <none>
web-nginx-748d764f7-4zvm6   1/1     Running   0          8m6s   10.244.0.4   minikube   <none>           <none>
web-nginx-748d764f7-fxb8r   1/1     Running   0          8m6s   10.244.0.5   minikube   <none>           <none>
web-nginx-748d764f7-j2hb9   1/1     Running   0          8m6s   10.244.0.3   minikube   <none>           <none>
web-nginx-748d764f7-prfdl   1/1     Running   0          3m     10.244.0.7   minikube   <none>           <none>
```